LESSON 01 / 05

Heine's criterion turns function limits into sequence limits

For a function defined near a,

$$\lim_{x\to a} f(x) = L$$

holds exactly when **every** sequence $x_n \to a$ with $x_n \neq a$ satisfies $f(x_n) \to L$. To prove nonexistence, two sequences with different output limits are enough. To prove existence, checking only a favourite sequence is not enough; the statement must cover every approach.

WORKED EXAMPLE

Two sequences disprove the limit of sin(1/x)

Take

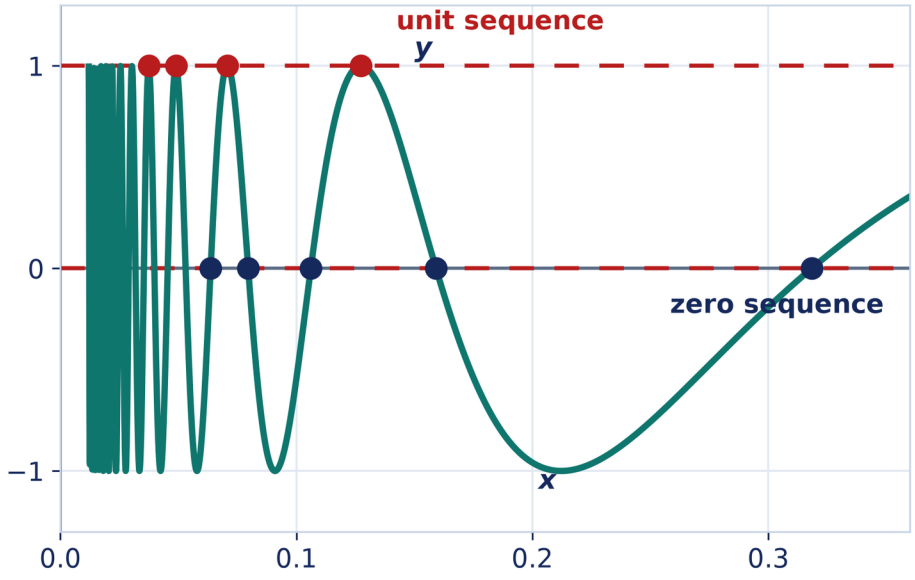
$$x_n = \frac{1}{n\pi}, \qquad y_n = \frac{1}{\pi/2 + 2n\pi}.$$

Both tend to 0 from the right, but

$$\sin\frac{1}{x_n} = \sin(n\pi) = 0, \qquad \sin\frac{1}{y_n} = 1.$$

Thus $\sin(1/x)$ has no limit as $x \to 0$. The oscillation is not a failure of algebra; it is verified nonexistence.

Two marked sequences approach x=0 but keep different heights



READ THE FIGURE

The zero-height points use x_n=1/(n pi); the unit-height points use y_n=1/(pi/2+2n pi). Their inputs all approach 0.

Dense sets can make every approach fail

Define $D(x) = 1$ for rational x and $D(x) = 0$ for irrational x. At any real a, rationals and irrationals are both dense, so there are sequences $q_n \to a$ and $r_n \to a$ with

$$D(q_n) = 1, \qquad D(r_n) = 0.$$

Therefore the Dirichlet function has no limit at any point. This is stronger than having a single jump or removable discontinuity.

A substitution can certify every sequence at once

For $x \neq 0$, put $u = 1/x^2$. Whenever $x_n \to 0$ with $x_n \neq 0$, we have $u_n = 1/x_n^2 \to \infty$, and hence

$$e^{-1/x_n^2} = e^{-u_n} \longrightarrow 0.$$

Heine's criterion now proves the two-sided limit $e^{-1/x^2} \to 0$. Assigning the value 0 at $x = 0$ therefore makes this function continuous there.

KEY RESULT

Existence and nonexistence need different amounts of evidence

One sequence can never prove a full function limit. One pair of sequences can disprove it. For a positive proof, use an estimate valid on a whole punctured neighbourhood or a substitution valid for every admissible sequence.

LESSON 02 / 05

Stolz–Cesàro is a discrete quotient rule

If b_n is strictly increasing and unbounded, and

$$\frac{a_{n+1}-a_n}{b_{n+1}-b_n} \longrightarrow L,$$

then $a_n/b_n \longrightarrow L$. The denominator increments are eventually positive, so the difference quotient controls successive weighted averages. Check the monotonicity and unboundedness before using the rule.

WORKED EXAMPLE

Arithmetic sums and harmonic growth

For $a_n = 1 + 2 + \cdots + n$ and $b_n = n^2$,

$$\frac{a_{n+1}-a_n}{b_{n+1}-b_n} = \frac{n+1}{2n+1} \longrightarrow \frac{1}{2},$$

so $(1 + \cdots + n)/n^2 \to 1/2$.

For $H_n = 1 + 1/2 + \cdots + 1/n$ and $b_n = \ln n$,

$$\frac{H_{n+1}-H_n}{\ln(n+1)-\ln n} = \frac{1/(n+1)}{\ln(1+1/n)} \longrightarrow 1.$$

Thus $H_n/\ln n \to 1$.

A recursive sequence needs convergence before algebra

If $a_{n+1} = f(a_n)$ with continuous f and $a_n \to L$, then passing to the limit gives $L = f(L)$. That equation is the **last** step, not a proof that the sequence converges. A common positive proof is monotone + bounded.

WORKED EXAMPLE

A nested-square-root recurrence

Let $a_1 = 1$ and $a_{n+1} = \sqrt{2 + a_n}$. If $1 \le a_n < 2$, then

$$a_{n+1} = \sqrt{2 + a_n} < 2.$$

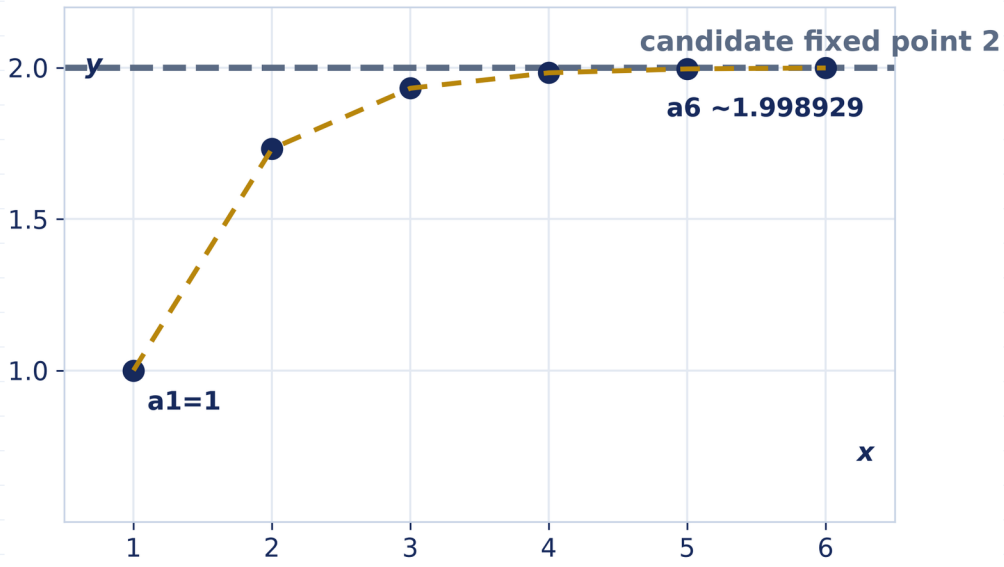
Also $a_{n+1} \ge a_n$ because $2 + a_n \ge a_n^2$ is equivalent to $(2 - a_n)(a_n + 1) \ge 0$. The base case $a_1 = 1$ starts the induction, so the sequence is increasing and bounded above by 2.

Therefore L exists and

$$L = \sqrt{2 + L} \quad \Longrightarrow \quad L^2 - L - 2 = 0.$$

The roots are 2 and -1; positivity selects $L = 2$.

The recurrence increases toward its fixed point 2



READ THE FIGURE
The dots are sequence terms, not a continuous function. Monotone increase and the upper bound prove convergence before the fixed-point equation is used.

WORKED EXAMPLE

Heron's iteration converges downward to sqrt(2)

Let $a_1 = 2$ and $a_{n+1} = (a_n + 2/a_n)/2$. AM–GM gives $a_{n+1} \geq \sqrt{2}$. When $a_n \geq \sqrt{2}$, we also have $2/a_n \leq a_n$, so $a_{n+1} \leq a_n$. Thus the sequence is decreasing and bounded below.

Its limit satisfies $L = (L + 2/L)/2$, so $L^2 = 2$. Positivity gives $L = \sqrt{2}$, not the algebraic alternative $-\sqrt{2}$.

WORKED EXAMPLE

An exponent-tracking recurrence

For $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2a_n}$, induction gives

$$a_n = 2^{1-2^{-n}} \longrightarrow 2.$$

Here an explicit exponent formula replaces the monotone-and-bounded argument.

COMMON TRAP

A fixed-point equation can have extraneous candidates

Writing $L = f(L)$ before proving convergence may produce numbers that the sequence never approaches. Establish monotonicity and a bound, or another convergence argument, then use continuity and discard candidates using sign and range information.

LESSON 03 / 05

Recognize a right-endpoint Riemann sum

For continuous f on $[0,1]$,

$$\frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right) \longrightarrow \int_0^1 f(x)\,dx.$$

The factor $1/n$ is the mesh and k/n are right endpoints. This extension previews the definite-integral theory developed fully in Chapter 21; here it is used as a sequence-limit technique.

WORKED EXAMPLE

Expose the sample point before integrating

$$\sum_{k=1}^{n} \frac{n}{n^2+k^2} = \frac{1}{n}\sum_{k=1}^{n} \frac{1}{1+(k/n)^2} \longrightarrow \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}.$$

Similarly,

$$\frac{1}{n}\sum_{k=1}^{n} \frac{1}{1+k/n} \longrightarrow \int_0^1 \frac{dx}{1+x} = \ln 2.$$

For $p>0$, factoring n^p gives

$$\frac{1^p+2^p+\cdots+n^p}{n^{p+1}} = \frac{1}{n}\sum_{k=1}^{n} \left(\frac{k}{n}\right)^p \longrightarrow \frac{1}{p+1}.$$

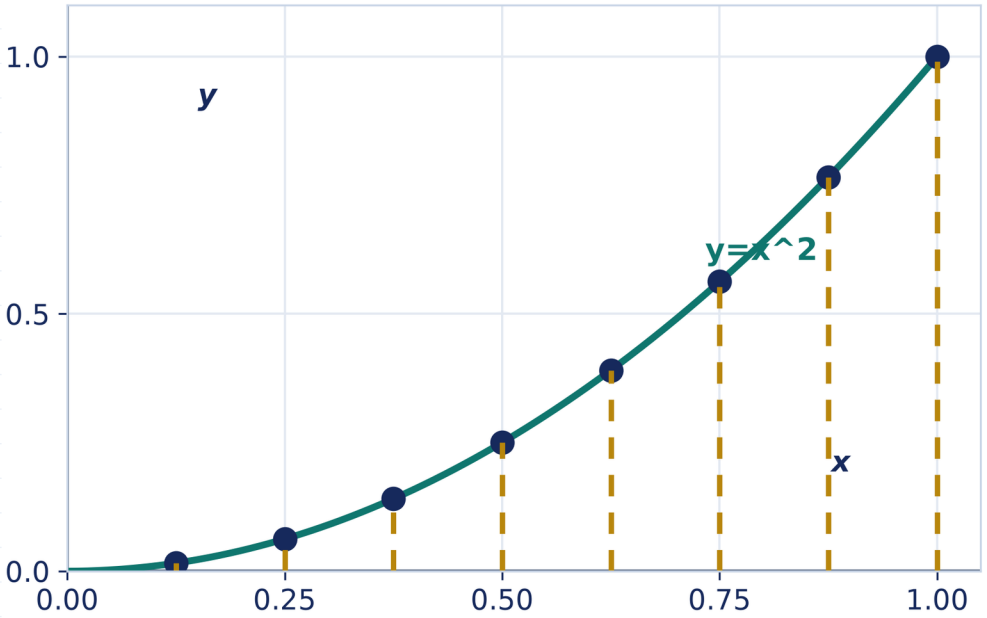
The values $p = 1/2, 1, 2, 3$ give $2/3, 1/2, 1/3, 1/4$ respectively.

KEY RESULT

Keep the mesh, sample point and error order visible

A sum is not a Riemann sum until the factor $1/n$ and the sampled function are explicit. A product is not evaluated by replacing every factor by 1: infinitely many small contributions can add to a nonzero logarithm.

Eight right endpoints approximate the area under y=x^2



READ THE FIGURE

For n=8 the right-endpoint sum is 51/128 = 0.3984375, above the limit 1/3. Increasing n refines the mesh; the samples are not an unweighted average.

Products become sums after logarithms

Let

$$P_n = \prod_{k=1}^{n} \left(1 + \tfrac{k}{n^2}\right).$$

Then $\ln P_n = \sum_{k=1}^{n} \ln(1 + k/n^2)$. For $x \ge 0$,

$$x - \tfrac{x^2}{2} \le \ln(1+x) \le x.$$

Summing the upper bound gives

$$\sum_{k=1}^{n} \tfrac{k}{n^2} = \tfrac{n+1}{2n} \longrightarrow \tfrac{1}{2}.$$

The subtracted error is bounded by

$$\tfrac{1}{2} \sum_{k=1}^{n} \tfrac{k^2}{n^4} = \tfrac{n(n+1)(2n+1)}{12n^4} \longrightarrow 0.$$

Hence $\ln P_n \to 1/2$, and continuity of the exponential gives $P_n \to \sqrt{e}$.

LESSON 04 / 05

Asymptotic equivalence preserves ratios, not differences

Write $u_n \sim v_n$ when $u_n/v_n \to 1$. Stirling's formula states

$$n! \sim \sqrt{2\pi n}\,\left(\frac{n}{e}\right)^n.$$

One proof compares $\ln(n!) = \sum_{k=1}^{n} \ln k$ with an integral and then tracks the correction terms. The formula is especially useful when factorial powers cancel in a ratio.

WORKED EXAMPLE

Central-binomial decay

Applying Stirling to numerator and denominator,

$$(2n)! \sim \sqrt{4\pi n}\,\left(\frac{2n}{e}\right)^{2n}, \qquad (n!)^2 \sim 2\pi n\left(\frac{n}{e}\right)^{2n}.$$

Therefore

$$\frac{(2n)!}{4^n (n!)^2} \sim \frac{1}{\sqrt{\pi n}}.$$

In particular the ratio tends to 0, while n times its square tends to $1/\pi$. Merely writing every factorial as "large" would lose this exact decay rate.

Harmonic sums separate a logarithm from a constant

The Euler–Mascheroni constant is defined by

$$H_n - \ln n \longrightarrow \gamma \approx 0.5772156649.$$

Consequently $H_n/\ln n \to 1$, but the difference $H_n - \ln n$ does **not** tend to 0. This illustrates why asymptotic equivalence cannot automatically be subtracted.

WORKED EXAMPLE

Cancel gamma without assuming it is zero

Since $\ln(2n) = \ln 2 + \ln n$,

$$H_{2n} - H_n = (H_{2n} - \ln(2n)) - (H_n - \ln n) + \ln 2 \longrightarrow \gamma - \gamma + \ln 2 = \ln 2.$$

Thus $H_{2n} - H_n - \ln 2 \to 0$, while $H_{2n} - H_n$ itself tends to $\ln 2$.

A second-order term determines the convergence rate

The notation $R_n = O(n^{-2})$ means $|R_n| \le Cn^{-2}$ for some constant C and all large n. Starting with

$$\ln(1+x) = x - \tfrac{x^2}{2} + O(x^3) \qquad (x \to 0),$$

put $x = 1/n$ and multiply by n:

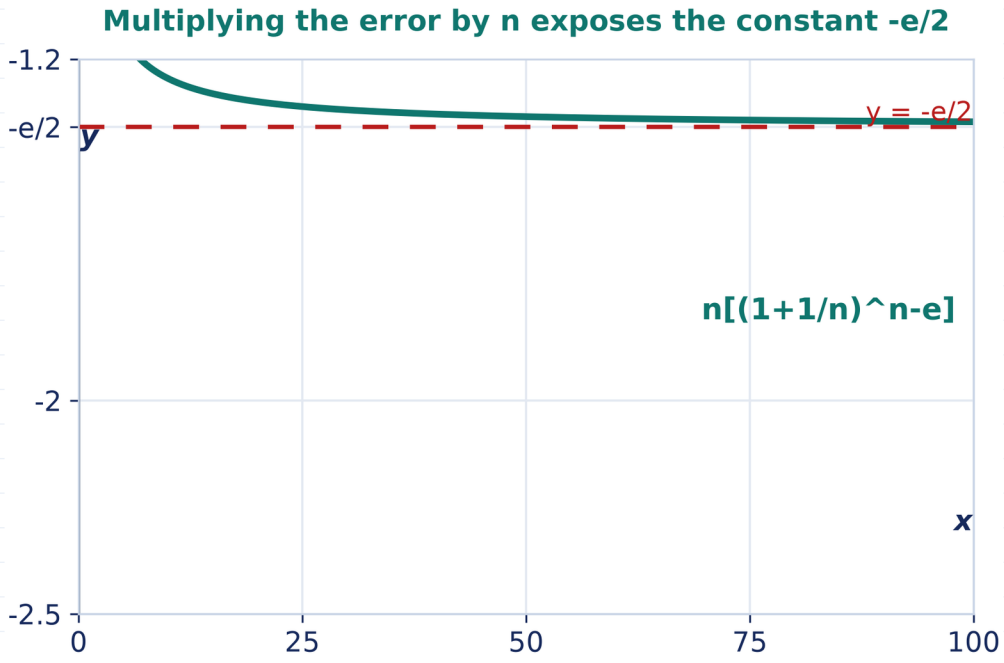
$$n\ln\left(1+\tfrac{1}{n}\right) = 1 - \tfrac{1}{2n} + O(n^{-2}).$$

Exponentiating and expanding $e^u = 1 + u + O(u^2)$ gives

$$\left(1+\tfrac{1}{n}\right)^n = e\left(1 - \tfrac{1}{2n} + O(n^{-2})\right).$$

Therefore

$$n\left[\left(1+\tfrac{1}{n}\right)^n - e\right] \longrightarrow -\tfrac{e}{2}.$$



READ THE FIGURE
The unscaled error tends to 0, but the scaled error tends to -e/2. A graph of the original sequence alone would hide this first-order rate.

COMMON TRAP
The leading limit and the rate answer different questions
The statements $(1+1/n)^n \to e$ and $n[(1+1/n)^n - e] \to -e/2$ are compatible. The second contains more information. To obtain it, expand one order beyond the first constant term and keep the Big-O remainder under control.

LESSON 05 / 05

Build a function whose endpoint signs are forced to oppose

Suppose f is continuous on $[0,1]$ and $f(0)=f(1)$. Define

$$g(x)=f(x)-f\left(x+\tfrac{1}{2}\right), \qquad 0\le x\le \tfrac{1}{2}.$$

Then

$$g\left(\tfrac{1}{2}\right)=f\left(\tfrac{1}{2}\right)-f(1)=f\left(\tfrac{1}{2}\right)-f(0)=-g(0).$$

If $g(0)=0$, take $c=0$. Otherwise IVT gives $c\in(0,1/2)$ with $g(c)=0$. Hence

$$f(c)=f\left(c+\tfrac{1}{2}\right).$$

WORKED EXAMPLE

An exact half-shift check

For $f(x)=(x-1/2)^2$, we have $f(0)=f(1)=1/4$ and

$$g(x)=\left(x-\tfrac{1}{2}\right)^2-x^2=\tfrac{1}{4}-x.$$

Thus $c=1/4$, and indeed $f(1/4)=f(3/4)=1/16$. The same antipodal-difference idea applied to continuous temperature T on a circle gives at least one pair of antipodal points with equal temperature.

Continuity extends rational information to all reals

An additive function satisfies $f(x+y)=f(x)+f(y)$. Induction and scaling give

$$f(qx)=qf(x) \qquad (q\in\mathbb{Q}).$$

If f is continuous and $a=f(1)$, then for any real x, choose rationals $q_k\to x$. Continuity gives

$$f(x)=\lim_{k\to\infty} f(q_k)=\lim_{k\to\infty} aq_k=ax.$$

Without continuity, additivity alone permits pathological solutions.

For the Jensen midpoint relation

$$f\left(\frac{x+y}{2}\right)=\frac{f(x)+f(y)}{2},$$

subtract $b=f(0)$. The function $g=f-b$ has the same midpoint property and $g(0)=0$. It follows that g is additive, so continuity at one point gives $g(x)=ax$ and therefore $f(x)=ax+b$.

WORKED EXAMPLE

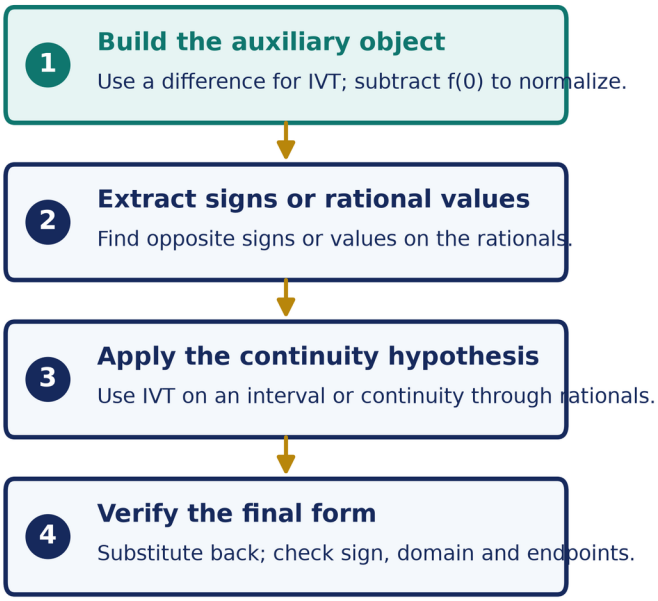
The continuous exponential functional equation

Suppose f is continuous, $f(x+y) = f(x)f(y)$, and $f(1) = a > 0$. First, $f(0) = 1$; the alternative $f(0) = 0$ would force $f \equiv 0$, contradicting $f(1) > 0$. Also $f(x) = f(x/2)^2 \geq 0$, and a zero value anywhere would force $f(1) = 0$, so f is strictly positive.

The functional equation gives $f(q) = a^q$ for every rational q. Rational density and continuity then extend this to

$$f(x) = a^x \qquad (x \in \mathbb{R}).$$

Continuity turns local or discrete information into a global conclusion



READ THE FIGURE

The algebra produces candidates; continuity is what makes the limiting or density argument legitimate.

KEY RESULT

Do not overstate antipodal conclusions

The argument guarantees ***at least one*** antipodal pair with equal values, not that every attained value occurs at antipodal pairs. Likewise, IVT guarantees a point with a required value, while a separate argument is needed for uniqueness or a formula for that point.